Task C1 — Sensor Simulator

Python simulator publishing to Mosquitto | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component C, Module 3

At a glance

Item	Detail
Deliverables	`simulator.py` (the simulator) and `test_simulator.py` (23 tests)
Required flag	`--anomaly {none|temp_drift|vibration|combined}`, default none
Broker	Mosquitto on `localhost:1883`, using the paho-mqtt library
Streams	Temperature 1 Hz, vibration RMS 0.5 Hz, door open/close events
Verification	All 23 tests pass; the MQTT path was tested against a paho stub, not a live broker (see Section 6)

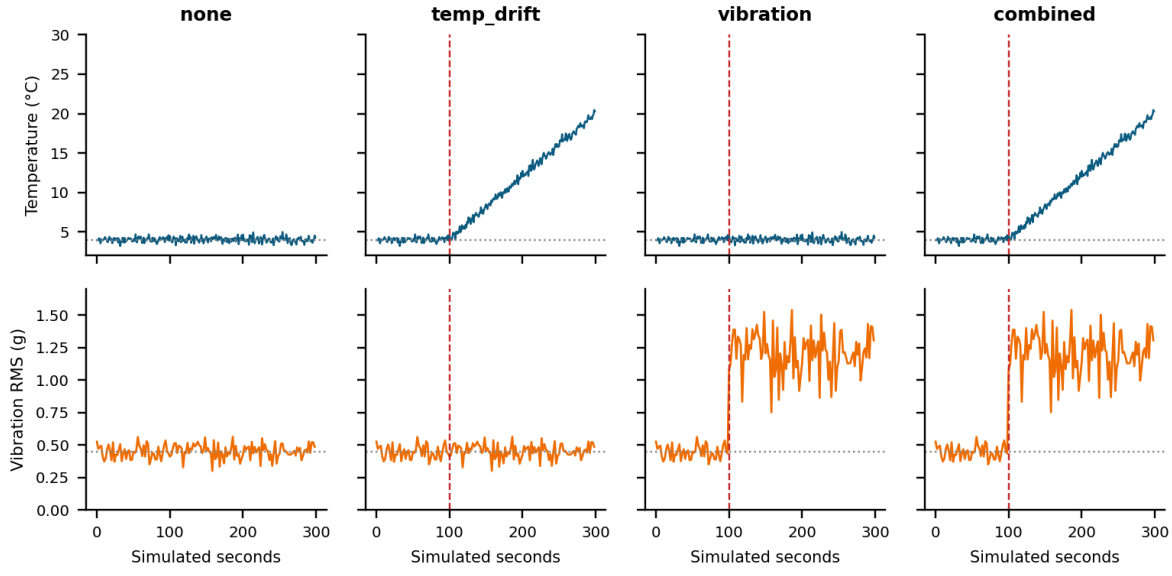
1 Stream design

Stream	Topic	Rate	QoS	Normal	Anomaly
temperature	truck/ID/temp	1 Hz	0	N(4.0 °C, 0.3); setpoint 4 °C	+0.08 °C per reading, linear drift
vibration_rms	truck/ID/vib	0.5 Hz	0	N(0.45 g, 0.05); compressor RMS	Step to N(1.2 g, 0.15); bearing wear
door_event	truck/ID/door	Discrete	1	OPEN or CLOSE with timestamp	Continues in every mode

Topic names and QoS levels follow the Task A2 design. A retained status message (online or offline, with an MQTT last will) is also published on truck/ID/status.

2 Anomaly modes

--anomaly	Temperature	Vibration	Door events
none	Normal	Normal	Normal
temp_drift	Drifts +0.08 °C per reading	Normal	Normal
vibration	Normal	Steps to N(1.2 g, 0.15)	Normal
combined	Drifts +0.08 °C per reading	Steps to N(1.2 g, 0.15)	Normal



Dashed red line: anomaly onset (--anomaly-start 100). Dotted grey line: normal mean (4.0 °C, 0.45 g).

Figure generated from the simulator's own output (seed 42, door events not plotted).

3 Message format

```
truck/T001/temp  qos=0  {"truck_id":"T001","ts":"2026-10-24T11:41:30.520Z","seq":1,"value":4.47,"unit":"C"}
truck/T001/vib   qos=0  {"truck_id":"T001","ts":"2026-10-24T11:41:30.520Z","seq":1,"value":0.567,"unit":"g_rms"}
truck/T001/door  qos=1  {"truck_id":"T001","ts":"2026-10-24T11:41:31.520Z","seq":1,"event":"OPEN"}
```

- Payloads are compact JSON of 83 B (temperature), 88 B (vibration) and 74 B (door event). They stay on the truck, so they do not affect the cellular figures in Task A1.
- Timestamps are UTC with millisecond precision, taken from the scheduled time, so readings are evenly spaced even if the operating system delays a send.
- **seq** counts messages per stream, so a subscriber can detect gaps.

4 Command-line reference

Flag	Default	Purpose
--anomaly	none	none, temp_drift, vibration or combined
--host, --port	localhost, 1883	Mosquitto broker address
--username, --password	none	Only if the broker requires authentication
--truck-id	T001	Identifier used in the topics
--duration	0	Simulated seconds to run; 0 runs until Ctrl-C
--speed	1.0	Simulated seconds per real second, for fast test runs
--seed	random	Makes the data reproducible
--anomaly-start	0	Simulated seconds of normal data before the anomaly begins
--door-interval	300	Mean seconds between door openings
--temp-cap	none	Optional temperature ceiling, e.g. 35 for ambient
--connect-timeout	10	Seconds to wait for the broker before exiting with an error
--dry-run, -v	off	Print messages instead of publishing; log every message

5 How to run it

```
# 1. Install the client library and start the broker
pip install paho-mqtt
mosquitto -v                          # or: sudo systemctl start mosquitto

# 2. In a second terminal, watch every topic for truck T001
mosquitto_sub -h localhost -t 'truck/T001/#' -v

# 3. In a third terminal, run the simulator
python3 simulator.py --anomaly none
python3 simulator.py --anomaly temp_drift --anomaly-start 60 --temp-cap 35
python3 simulator.py --anomaly vibration --seed 1
python3 simulator.py --anomaly combined --duration 600 --speed 10

# 4. Run the tests (no broker needed)
python3 -m unittest -v test_simulator
```

6 Verification

Check	What it confirms	Result
Sensor statistics	Normal temperature and vibration match N(4.0, 0.3) and N(0.45, 0.05); fault vibration matches N(1.2, 0.15)	Pass
Linear drift	Fitted slope is 0.080 °C per reading (tolerance 0.002)	Pass
Rates and spacing	100 temperature and 50 vibration messages per 100 s, spaced exactly 1 s and 2 s	Pass
Anomaly modes	none, temp_drift, vibration and combined each behave as specified; onset delay works	Pass
Door events	Start with OPEN, strictly alternate, open for 20–120 s, continue in every mode	Pass
Reproducibility	Same seed gives identical data; a different seed differs	Pass
CLI	Defaults correct; all four modes accepted; invalid mode rejected with exit code 2	Pass
MQTT adapter (paho stub)	Connects to localhost:1883, sets last will, publishes correct topics, QoS and retain flags, sends online and offline status, disconnects cleanly; exits with code 1 if no broker	Pass
Real-time run	10 s run took 10.09 s and sent 10 temperature and 5 vibration messages	Pass
Ctrl-C	Stops cleanly with exit code 0 and a summary line	Pass
Live Mosquitto	Not run: paho-mqtt and Mosquitto are not available in my environment	Please run Section 5

Observed over 3,600 s (seed 42)	Temperature	Vibration RMS
none	mean 4.00 °C, sd 0.30	mean 0.451 g, sd 0.050
vibration	mean 4.00 °C, sd 0.30	mean 1.202 g, sd 0.150
temp_drift	slope 0.0800 °C per reading, residual sd 0.301	mean 0.451 g, sd 0.050

7 Design notes and items to confirm

- **Drift is fast.** 0.08 °C per reading at 1 Hz is 4.8 °C per minute, 4.8× the 1 °C per minute used in the Task A1 latency analysis. Cargo reaches +1 °C after about 13 s and +3 °C after about 38 s. Without a cap it passes a 35 °C ambient after about 6.5 minutes (31 ÷ 0.08 = 388 readings), so use `--temp-cap 35` for long runs.
- **Vibration is RMS at 0.5 Hz**, one value every 2 s, as in the table. Task A2 described raw 500 Hz blocks on the same topic. I kept the topic name `vib`; tell me if you prefer `vib_rms`.
- **Door events in anomaly modes.** I read the table's last row as: door events continue unchanged in every mode, and “combined” means both anomalies together. Please confirm.
- **Door frequency.** The default mean gap is 300 s so short runs show events. Task A1 assumed 50 events per day, which is about `--door-interval 3400`.
- **Anomaly onset.** It starts at time zero by default; use `--anomaly-start` for a normal period first.
- **Broker details.** Confirm whether your Mosquitto needs a username and password, and which paho-mqtt version you use (1.x and 2.x are both supported).